Physics 5020: quiz 8. Student's name:

(Dated: March 2, 2011)

Problem 1 [1 point].

A segment of wire is bent into an arc of radius R and subtended angle θ, as shown in the figure below. Point P is at the center of the circular segment. The wire carries current I. What is the magnitude of the magnetic field at P ? (A) 0, (B) $\frac{\mu_0 I\theta}{(2\pi)^2 R}$, (C) $\frac{\mu_0 I\theta}{4\pi R}$, (D) $\frac{\mu_0 I\theta}{4\pi R^2}$, (E) $\frac{\mu_0 I}{2\theta R^2}$

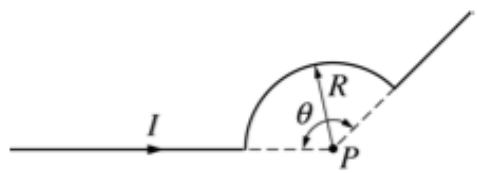
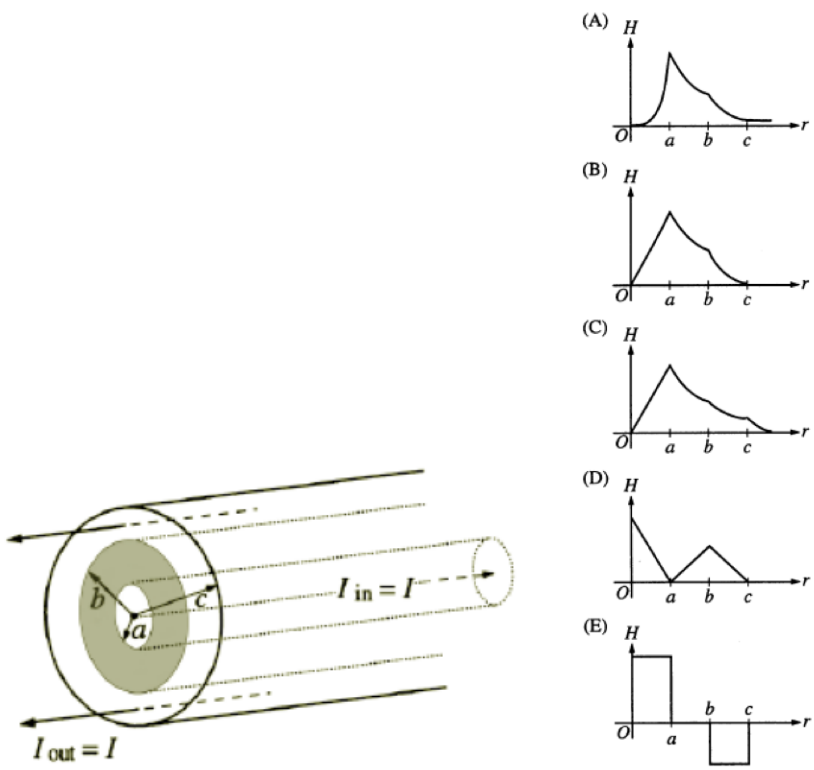


FIG. 1: The bent wire.

Problem 2 [2 points].

A coaxial cable has the cross section shown in Fig.2(a) below. The shaded region is insulated. The regions in which $r < a$ and $b < r < c$ are conducting. A uniform dc current density of total current I flows along the inner part of the cable ($r < a$) and returns along the outer part of the cable ($b < r < c$) in the directions shown. The radial dependence of the magnitude of the magnetic field, H, is shown by which of the following in Fig.2(b) below?



(a) (b)

FIG. 2: Problem 2. (a) Coaxial cable. (b) Magnetic field vs radial distance r dependence.

Problem 3 [2 points].

Two long conductors are arranged as shown in Fig.3 to form overlapping cylinders, each of radius r, whose centers are separated by a distance d. Current density J flows into the plane of the page along the shaded part of one conductor and an equal current flows out of the plane of the page along the shaded portion of the other, as shown. What are the magnitude and direction of the magnetic field at point A?

(A) $(\mu_0/2\pi)\pi dJ$, in the $+y$-direction, (B) $(\mu_0/2\pi)d^2J/r$, in the $+y$-direction, (C) $(\mu_0/2\pi)4d^2J/r$, in the $-y$-direction, (D) $(\mu_0/2\pi)Jr^2/d$, in the $-y$-direction, (E) there is no magnetic field at A.

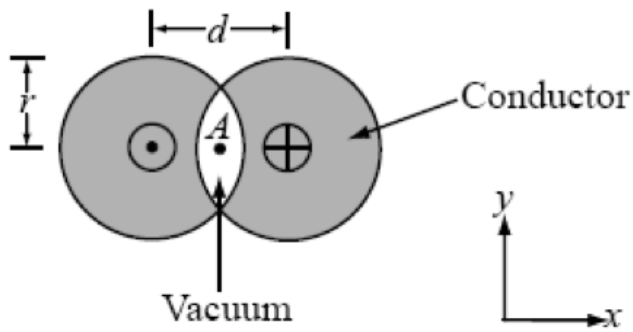


FIG. 3: The set-up.